

MUSEUM METADATA ANALYSIS

Predicting Artwork Centuries

Chronological Classification via Machine Learning

PROJECT OVERVIEW



Data Mining

Extraction of 15,078 raw records from **AIC** and **The Met** APIs. Final cleaned dataset consists of 8,929 high-fidelity samples across 10 centuries.



Engineering

Automated date normalization and century labeling. Text vectorization using **TF-IDF** (Sparse) and **Doc2Vec** (Dense) methodologies.



Modeling

Comparison of Logistic Regression, k-NN, and **Random Forest**. The latter emerged as the top performer with a baseline accuracy of **78.9%**.



Analysis

Interpretation of curatorial signals through Feature Importance and high-dimensional visualization via **t-SNE** clustering.

DATASET DISTRIBUTION

7,143

Training Samples

1,786

Testing Samples

Data Integrity

The model was trained on a strictly curated subset of the AIC and Met collections.

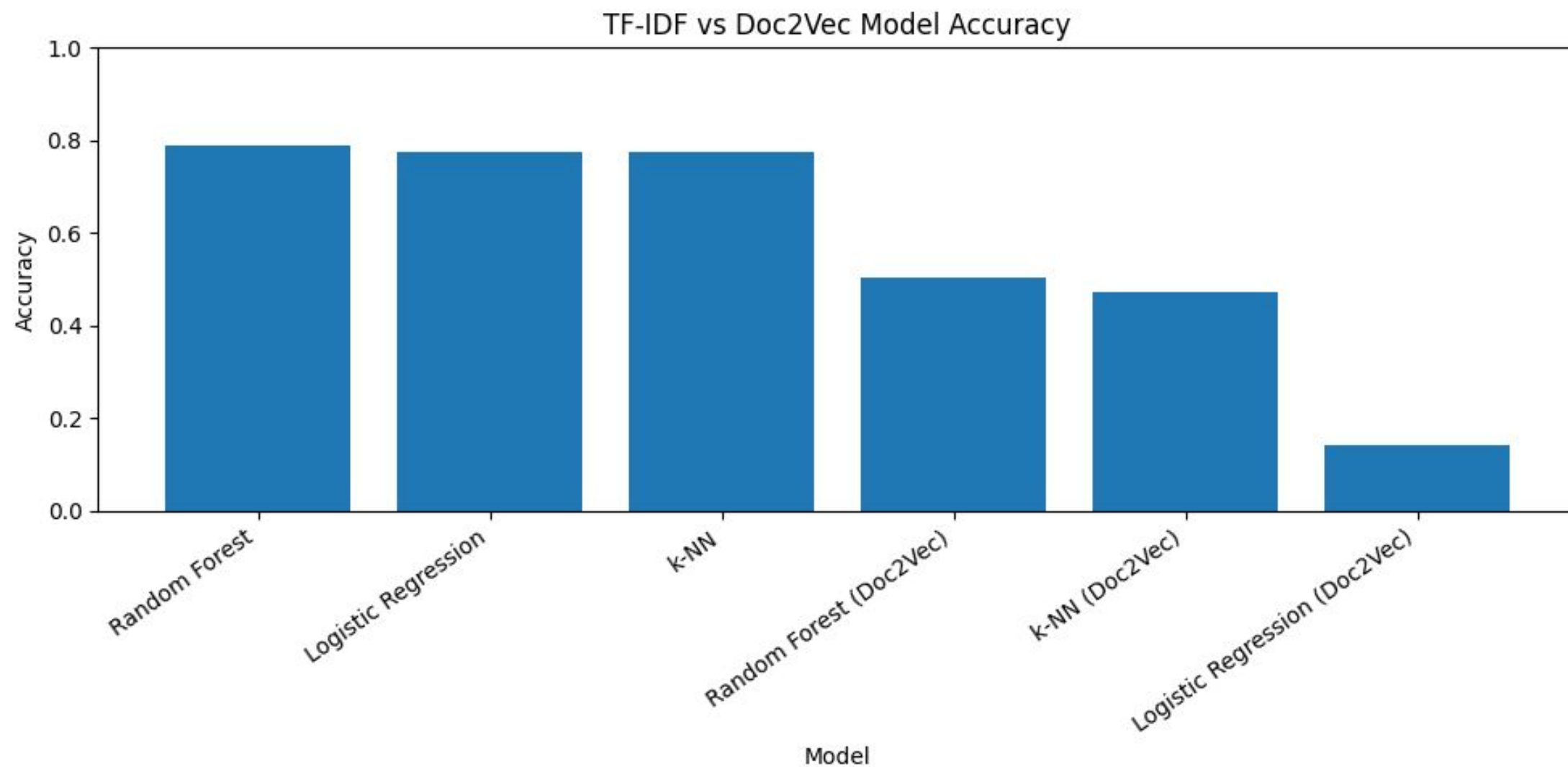
- **Combined Raw Samples:** 15,078 records.
- **Model Dataset:** 8,929 samples (Top 10 centuries).
- **80/20 Split:** to ensure representative century distribution in both sets.

BENCHMARKING PERFORMANCE



Random Forest achieved the highest overall accuracy. Logistic Regression excelled in specific minority classes where high-coefficient "anchor words" were present. Furthermore, Random Forest achieved higher accuracy because its ensemble architecture successfully captured complex, non-linear interactions between specific curatorial keywords in the metadata while providing robustness against the high-dimensional, sparse nature of the TF-IDF vectorized dataset.

PERFORMANCE between TF-IDF and DOC2VEC



WHY TF-IDF OUTPERFORMS DOC2VEC

Codified Metadata

Museum cataloging is keyword-intensive. Terms like "Tempera" or "Acrylic" are binary temporal markers. TF-IDF preserves these signals, Doc2Vec attempts to find broad semantic context. However, in artwork metadata, a single word like "born" carries more temporal weight than a latent "meaning" vector.

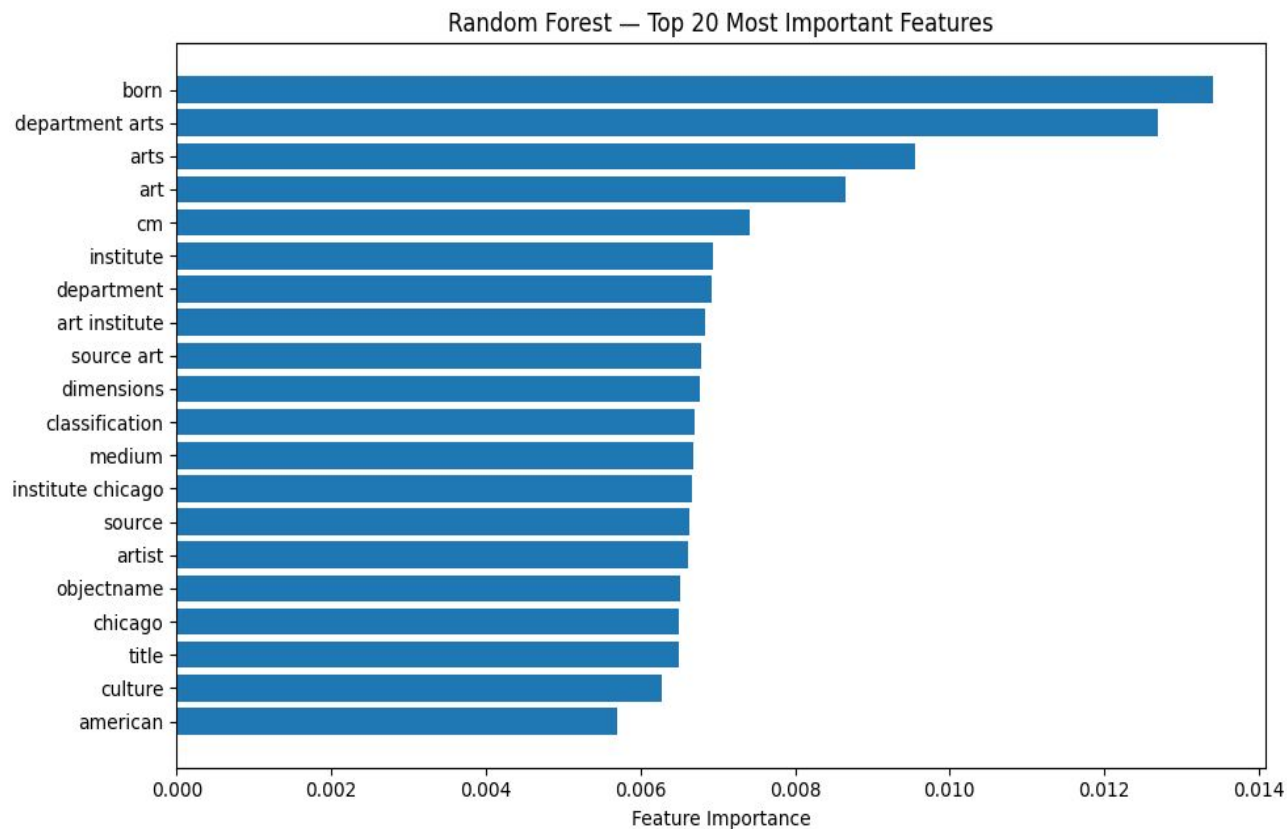
Feature Sparsity

Metadata strings are relatively short. TF-IDF excels at highlighting rare, high-information tokens (e.g., "Impressionist") which act as absolute chronological proxies.

Interpretability

TF-IDF allows for direct feature weight analysis. Understanding why a model chose the 21st century (e.g., the word "born") is critical for curatorial validation.

FEATURE IMPORTANCE ANALYSIS

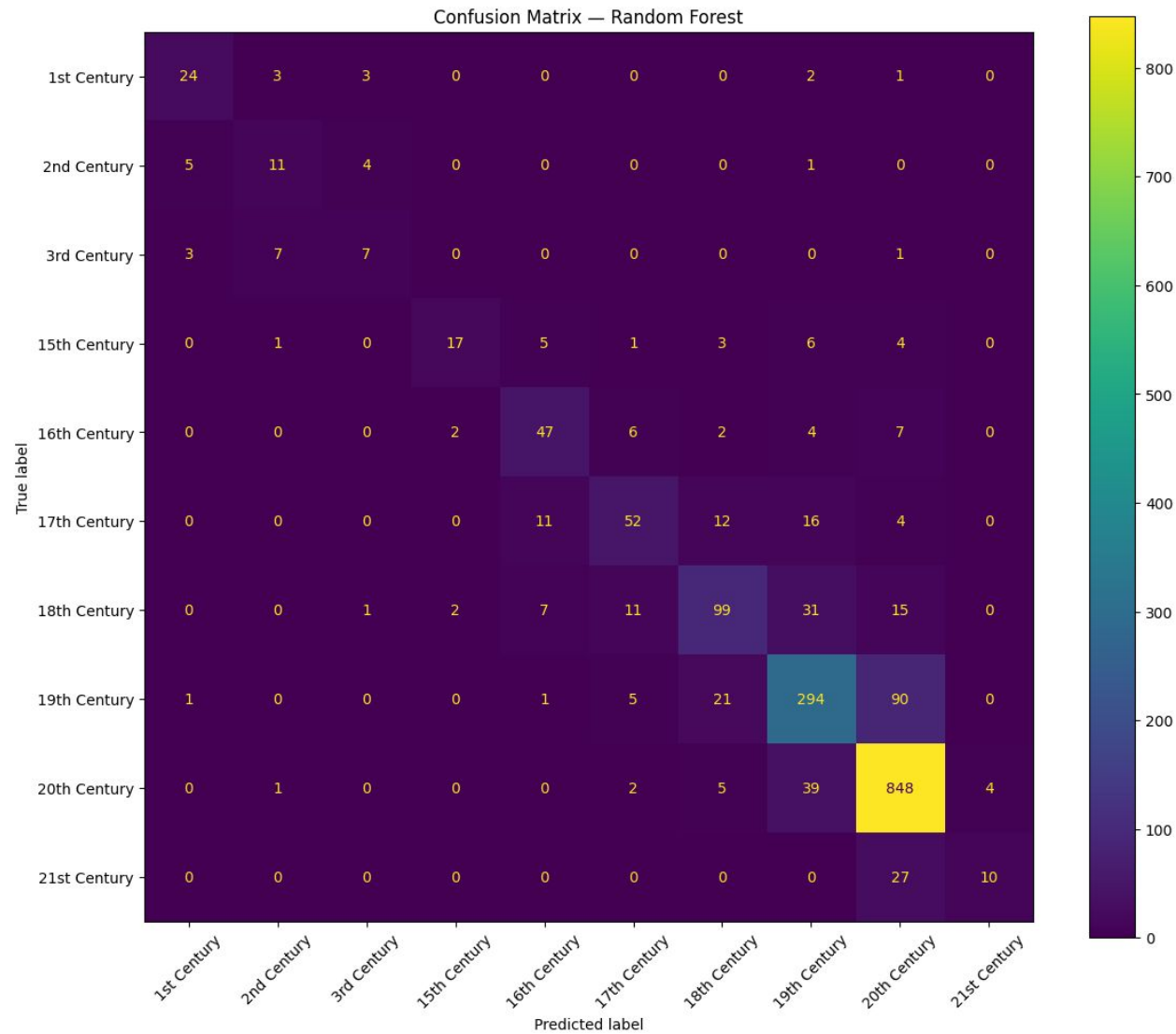


Key Chronological Signals

Top features reveal the model's reliance on specific curatorial metadata:

- '**Born**': The most important feature for 21st-century classification, identifying living artist bio-strings.
- '**Department**': Acts as a coarse-grain filter (e.g., "Modern Art").
- '**Tempera/Panel**': Highly indicative of the 15th-16th centuries.

CONFUSION MATRIX



Analysis

Our analysis of the confusion matrix reveals that errors are rarely random; they are almost exclusively "off-by-one" errors.

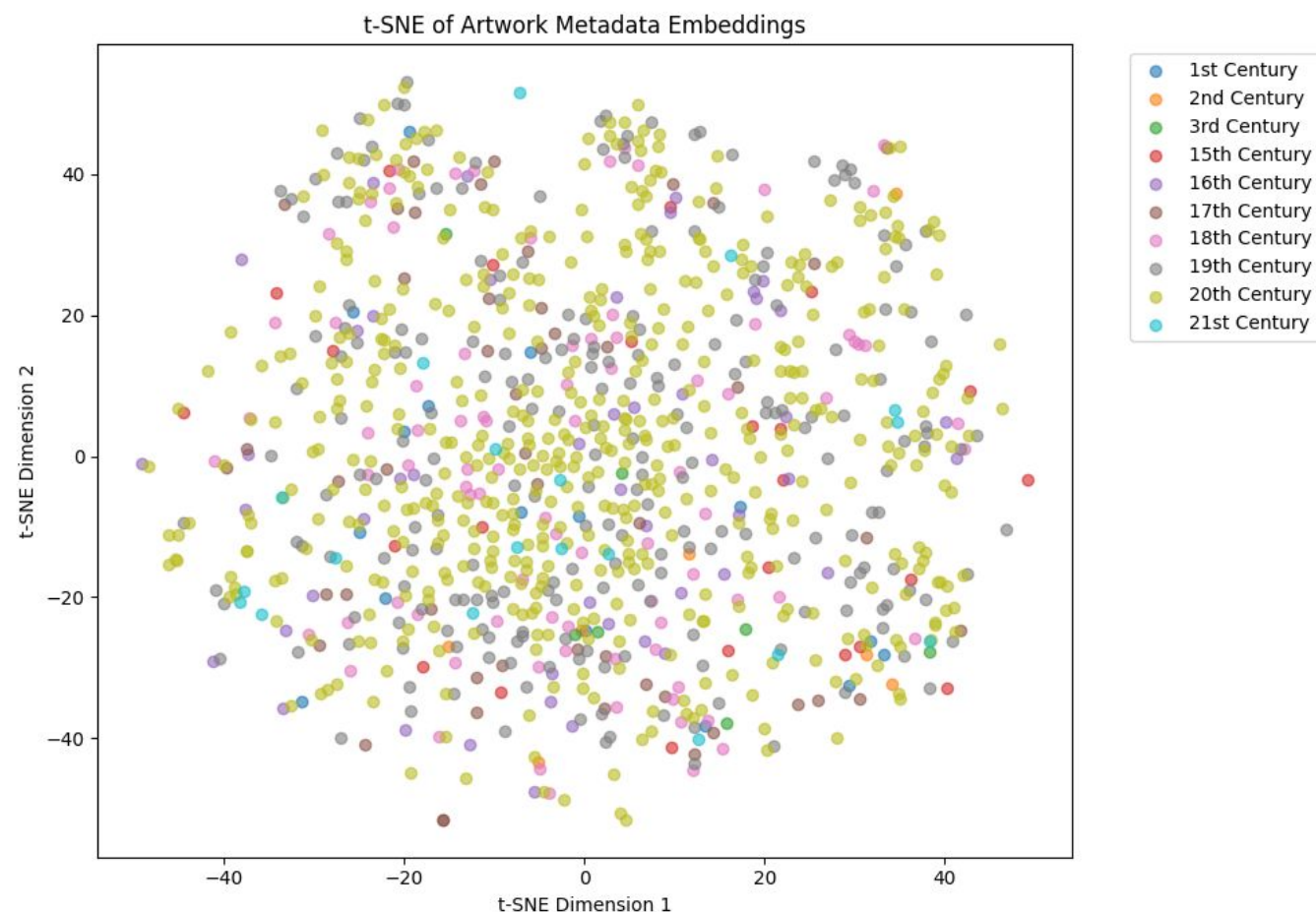
- **Adjacency:** 19th-century works are most often misclassified as 20th-century.
- **Transitional Styles:** Late 18th-century "Neoclassical" art shares metadata tokens with early 19th-century works.
- **Conclusion:** The model understands temporal flow, with failures occurring at stylistic border points.. The model is extremely biased towards the 29th century as there are more samples for that century.

VISUALIZING SEMANTIC CLUSTERS

t-SNE Visualization

Projecting Doc2Vec embeddings into 2D space reveals structural patterns in the collection:

- **Modern Core:** A dense cluster representing the massive 20th-century collection.
- **Ancient Outliers:** Distinct islands for Greek and Roman antiquities.
- **Overlaps:** The high overlap in 18th-19th century works explains the classification drop in Doc2Vec models compared to TF-IDF.



Questions?